Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division C | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) One mm over one km^2 equals…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. 1 m^3

B. 1 million m^3

C. 1000 liters

D. 1000 m^3

Correct: D

2. (2 points) Dissolved load consists of…

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Only gravel

B. Only floating wood

C. Only sand

D. Ions and molecules in solution

Correct: D

3. (2 points) High porosity always means high permeability?

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Always

B. Only for clay

C. Only below water table

D. No

Correct: D

4. (2 points) A budget requires…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. Only a river name

B. Only a map color

C. Only a starting temperature

D. A boundary and interval

Correct: D

5. (2 points) Competence describes…

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Total water volume

B. Only sediment color

C. Only rainfall

D. Largest transportable particles

Correct: D

6. (2 points) Darcy flux equals…

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Q x A

B. K/A

C. Porosity alone

D. Q/A

Correct: D

7. (2 points) Infiltrated water can later…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. Never move again

B. Only evaporate instantly

C. Create salt by definition

D. Support streamflow

Correct: D

8. (2 points) Equal discharge guarantees equal velocity?

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Always

B. Only for rivers

C. Only at flood stage

D. No

Correct: D

9. (2 points) Flow generally follows…

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Only surface slope

B. Increasing head always

C. Only north

D. Decreasing hydraulic head

Correct: D

10. (4 points) What is missing from a simple P-Q budget?

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

answer: Evapotranspiration, storage change, and other boundary transfers may matter. (4 points).

11. (2 points) A freshwater catchment receives 110 mm precipitation, loses 40 mm to evapotranspiration and exports 30 mm runoff. Find storage change.

answer: 40 mm (2 points).

Numerical target 40; absolute tolerance 0.08. Accepted units: , mm.

12. (2 points) A rectangular stream is 5 m wide and 2 m deep with mean velocity 0.5 m/s. Calculate discharge.

answer: 5 m^3/s (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , m^3/s.

13. (2 points) A river channel is 13 km long between points 9 km apart along the valley. Find sinuosity.

answer: 1.4444444444444444 (2 points).

Numerical target 1.4444444; absolute tolerance 0.0028888889. Accepted units: , .

14. (2 points) A water table drops 2 m over 100 m horizontal distance. Find hydraulic gradient.

answer: 0.02 (2 points).

Numerical target 0.02; absolute tolerance 4e-05. Accepted units: , .

15. (2 points) Use Q=K A i with K=0.001 m/s, A=21 m^2 and i=0.02. Find saturated Darcy discharge.

answer: 0.00042 m^3/s (2 points).

Numerical target 0.00042; absolute tolerance 8.4e-07. Accepted units: , m^3/s.

16. (2 points) A lake stores 1000 m^3 and exports 100 m^3/day in a steady well-mixed teaching model. Estimate residence time as storage/outflow.

answer: 10 days (2 points).

Numerical target 10; absolute tolerance 0.02. Accepted units: , days.

17. (2 points) An urbanized watershed has more impervious area. Predict one hydrograph change under a comparable storm and explain.

answer: Faster runoff routing can shorten lag time and increase peak discharge; drainage infrastructure, storage and storm properties must be controlled. (2 points).

18. (2 points) A stream bends through its floodplain. Locate likely erosion and deposition relative to the bend.

answer: Erosion is commonly concentrated on the outer cut bank and deposition on the inner point bar, with local flow conditions controlling details. (2 points).

19. (2 points) Two first-order channels join, and no higher-order channel enters. What Strahler order results, and why?

answer: Second order; joining two channels of the same order increases order by one. (2 points).

20. (2 points) Groundwater pumping increases near a connected river. Explain a possible effect on river baseflow.

answer: Lower groundwater head can reduce discharge to the river or induce infiltration from it, depending on hydraulic connection and gradients. (2 points).